ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
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PHYSICS

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Modern Physics-2

English Medium

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

A. Binding energy of nucleus, nuclear fission & fusion

1. In Rutherford's famous gold foil scattering experiment, he found that most alpha particles would pass through the foil undeflected. Which one of the following nuclear properties can be inferred from this observation?

(A) The nucleus must have a positive charge

(B) Most of the mass of an atom is in the nucleus

(C) The nucleus contains both protons neutrons

(D) The diameter of the nucleus is small compared to the diameter of the atom

(E) None of these

MP0207

2. Let u be denote one atomic mass unit. One atom of an element of mass number A has mass exactly equal to Au

(A) for any value of A

(B) only for A = 1

(C) only for A = 12

(D) for any value of A provided the atom is stable

MP0208

3. If radius of the $^{27}_{13}Al$ nucleus is estimated to be 3.6 fermi, then the radius of $^{125}_{52}Te$ nucleus be nearly-

[AIEEE - 2005]

(A) 6 fermi (B) 8 fermi (C) 4 fermi (D) 5 fermi

MP0209

4. The surface area of a nucleus varies with mass number A as

(A) $A^{2/3}$ (B) $A^{1/3}$ (C) A (D) None

MP0210

5. A nucleus disintegrates into two nuclear parts which have their velocities in the ratio 2 : 1 The ratio of their nuclear sizes will be- **[AIEEE - 2004]**

(A) $2^{1/3} : 1$ (B) $1 : 3^{1/2}$ (C) $3^{1/2} : 1$ (D) $1 : 2^{1/3}$

MP0211

6. The binding energy per nucleon for C^{12} is 7.68 MeV and that for C^{13} is 7.5 MeV. The energy required to remove a neutron from C^{13} is

(A) 5.34 MeV (B) 5.5 MeV

(C) 9.5 MeV (D) 9.34 MeV

MP0212

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7. The following nuclear reaction is an example of ${}^{12}_{6}C + {}^{4}_{2}H \rightarrow {}^{16}_{8}O$ + energy

(A) fission (B) fusion
(C) alpha decay (D) beta decay

MP0213

8. Fast neutrons may most easily be slowed down by which one of the following methods?
(A) passing them through a substance rich in hydrogen
(B) allowing them to collide elastically with heavy nuclei
(C) using lead shielding
(D) passing them through an increasing potential gradient space

MP0214

9. A nuclear transformation is denoted by $X(n, \alpha) \rightarrow {}^{7}_{3}Li$. Which of the following is the nucleus of element X ? **[AIEEE - 2005]**

(A) ${}^{12}_{6}C$ (B) ${}^{10}_{5}B$ (C) ${}^{9}_{5}B$ (D) ${}^{11}_{4}Be$

MP0215

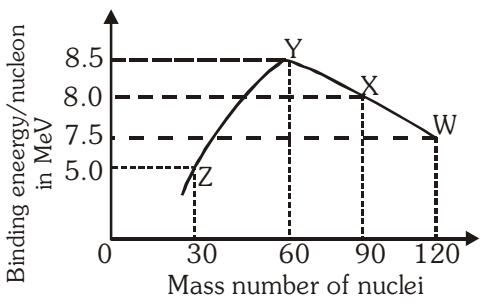
10. A certain radioactive nuclide of mass number m_x disintegrates, with the emission of an electron and γ radiation only, to give second nuclide of mass number m_y. Which one of the following equation correctly relates m_x and m_y?
(A) $m_y = m_x + 1$ (B) $m_y = m_x - 2$ (C) $m_y = m_x - 1$ (D) $m_y = m_x$

MP0216

11. If a star converts all of its Helium into oxygen nucleus, find the amount of energy released per nucleus of oxygen. O = 15.9994 amu and He = 4.0026 amu **[JEE' 2005 (Scr)]**
(A) 7.26 MeV (B) 7 MeV
(C) 10.24 MeV (D) 5.12 MeV

MP0220

12. Binding energy per nucleon versus mass number curve for nuclei is shown in figure. W, X, Y and Z are four nuclei indicated on the curve. The process that would release energy is :



(A) $Y \rightarrow 2Z$ (B) $W \rightarrow X + Z$ (C) $W \rightarrow 2Y$ (D) $X \rightarrow Y + Z$

MP0223

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13. An alpha nucleus of energy $\frac{1}{2}mv^2$ bombards a heavy nuclear target of charge Ze. Then the distance of closest approach for the alpha nucleus will be proportional to- **[AIEEE - 2006]**

(A) v^2 (B) $1/m$ (C) $1/v^4$ (D) $1/Ze$

MP0224

14. If the binding energy per nucleon in ^7_3Li and ^4_2He nuclei are 5.60 MeV and 7.06 MeV respectively, then in the reaction : $p + {}^7_3Li \rightarrow 2{}^4_2He$ energy of proton must be- **[AIEEE - 2006]**

(A) 28.24 MeV (B) 17.28 MeV
(C) 1.46 MeV (D) 39.2 MeV

MP0225

15. If M_0 is the mass of an oxygen isotope $_8O^{17}$, M_p and M_n are the masses of a proton and a neutron, respectively, the nuclear binding energy of the isotope is- **[AIEEE - 2007]**

(A) $(M_o - 8M_p)c^2$ (B) $(M_o - 8Mp - 9M_n)c^2$
(C) M_oc^2 (D) $(M_o - 17M_n)c^2$

MP0226

16. In the options given below, let E denote the rest mass energy of a nucleus and *n* a neutron. The correct option is :- **[JEE 2007]**

(A) $E\left(^{236}_{92}U\right) > E\left(^{137}_{53}I\right) + E\left(^{97}_{39}Y\right) + 2E(n)$

(B) $E\left(^{236}_{92}U\right) < E\left(^{137}_{53}I\right) + E\left(^{97}_{39}Y\right) + 2E(n)$

(C) $E\left(^{236}_{92}U\right) < E\left(^{140}_{56}Ba\right) + E\left(^{94}_{36}Kr\right) + 2E(n)$

(D) $E\left(^{236}_{92}U\right) = E\left(^{140}_{56}Ba\right) + E\left(^{94}_{36}Kr\right) + 2E(n)$

MP0227

17. When $_3Li^7$ nuclei are bombarded by protons and the resultant nuclei are $_4Be^8$, the emitted particles will be- **[AIEEE - 2006]**

(A) alpha particles (B) beta particles
(C) gamma photons (D) neutrons

MP0241

18. After absorbing a slowly moving neutron of mass m_N (momentum ~0) a nucleus of mass M breaks into two nuclei of masses m_1 and $5m_1 (6m_1 = M + m_N)$, respectively. If the de Broglie wavelength of the nucleus with mass m_1 is λ, then de Broglie wavelength of the other nucleus will be:- **[AIEEE - 2011]**

(A) 25λ (B) 5λ (C) $\frac{\lambda}{5}$ (D) λ

MP0282

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B. α, β, γ, decay

19. A nucleus with Z = 92 emits the following in a sequence : α, α, β^-, β^-, α, α, α, α, β^-, β^-, α, β^+, β^+, α. The Z of the resulting nucleus is- **[AIEEE - 2003]**

(A) 76 (B) 78 (C) 82 (D) 74

MP0217

20. In an α-decay the Kinetic energy of α particle is 48 MeV and Q-value of the reaction is 50 MeV. The mass number of the mother nucleus is: (Assume that daughter nucleus is in ground state)

(A) 96 (B) 100 (C) 104 (D) none of these

MP0218

21. When U^{238} nucleus originally at rest, decays by emitting an alpha particle having a speed u, the recoil speed of the residual nucleus is- **[AIEEE - 2003]**

(A) $\frac{4u}{238}$ (B) $-\frac{4u}{234}$ (C) $\frac{4u}{234}$ (D) $-\frac{4u}{238}$

MP0219

22. The nucleus of element X (A = 220) undergoes α-decay. If Q-value of the reaction is 5.5 MeV, then the kinetic energy of α-particle is : **[JEE 2003 (Scr)]**

(A) 5.4 MeV (B) 10.8 MeV (C) 2.7 MeV (D) None

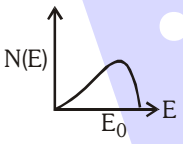
MP0221

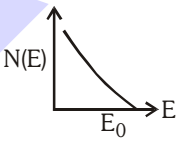
23. In the uranium radioactive series the initial nucleus is $_{92}U^{238}$, and the final nucleus is $_{82}Pb^{206}$. When the uranium nucleus decays to lead, the number of α – particles emitted is.. and the number of β-particles emitted...

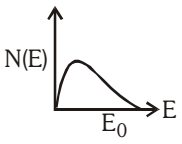
(A) 6, 8 (B) 8, 6 (C) 16, 6 (D) 32, 12

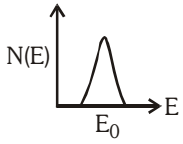
MP0222

24. The energy spectrum of β-particles [number N(E) as a function of β-energy E] emitted from a radioactive source is- **[AIEEE - 2006]**

(A) 

(B) 

(C) 

(D) 

MP0242

25. In gamma ray emission from a nucleus **[AIEEE-2007]**

(A) both the neutron number and the proton number change

(B) there is no change in the proton number and the neutron number

(C) only the neutron number changes

(D) only the proton number changes

MP0243

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26. **Statement-1:** A nucleus having energy E_1 decays be β^- emission to daughter nucleus having energy E_2, but the β^- rays are emitted with a continuous energy spectrum having end point energy $E_1 - E_2$.

Statement-1: To conserve energy and momentum in β-decay at least three particles must take part in the transformation. **[AIEEE - 2011]**

(A) Statement-1 is incorrect, statement-2 is correct

(B) Statement-1 is correct, statement-2 is incorrect

(C) Statement-1 is correct, statement-2 correct; statement-2 is the correct explanation of statement-1

(D) Statement-1 is correct, statement-2 is correct; statement -2 is not the correct explanation of statement-1.

MP0283

27. What is the essential distinction between X–rays and γ–rays

(A) γ–rays have shorter wavelength than X–rays

(B) γ–rays are extraterrestrial, X–rays are man–made

(C) γ–rays have less penetrating power than X–rays

(D) γ–rays originate from within an atomic nucleus, X–rays from outside an atomic nucleus.

MP0088

28. The intensity of gamma radiation from a given source is I. On passing through 36 mm of lead, it is reduced to I/8. The thickness of lead, which will reduce the intensity to I/2 will be- **[AIEEE-2005]**

(A) 6 mm (B) 9 mm (C) 18 mm (D) 12 mm

MP0090

C. Radioactivity

29. The half–life of ^{131}I is 8 days. Given a sample of ^{131}I at time t = 0, we can assert that :

(A) no nucleus will decay before t = 4 days

(B) no nucleus will decay before t = 8 days

(C) all nuclei will decay before t = 16 days

(D) a given nucleus may decay at any time after t = 0

MP0228

30. Activity of a radioactive substance is R_1 at time t_1 and R_2 at time $t_2 (t_2 > t_1)$. Then the ratio $\frac{R_2}{R_1}$ is:

(A) $\frac{t_2}{t_1}$ (B) $e^{-\lambda(t_1+t_2)}$ (C) $e\left(\frac{t_1 - t_2}{\lambda}\right)$ (D) $e^{\lambda(t_1-t_2)}$

MP0229

31. A particular nucleus in a large population of identical radioactive nuclei did survive 5 half lives of that isotope. Then the probability that this surviving nucleus will survive the next half life :

(A) $\frac{1}{32}$ (B) $\frac{1}{5}$ (C) $\frac{1}{2}$ (D) $\frac{1}{10}$

MP0230

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32. A certain radio active substance has a half life of 5 years. Thus for a nucleus in a sample of the element, the probability of decay in ten years is

(A) 50% (B) 75% (C) 100% (D) 60%

MP0231

33. The activity of a sample reduces from A_0 to $A_0/\sqrt{3}$ in one hour. The activity after 3 hours more will be :-

(A) $\frac{A_0}{3\sqrt{3}}$ (B) $\frac{A_0}{9}$ (C) $\frac{A_0}{9\sqrt{3}}$ (D) $\frac{A_0}{27}$

MP0232

34. Half life of radium is 1620 years. How many radium nuclei decay in 5 hours in 5 gm radium? (Atomic weight of radium = 223)

(A) 9.1×10^{12} (B) 3.23×10^{15}

(C) 1.72×10^{20} (D) 3.3×10^{17}

MP0233

35. The activity of a sample of radioactive material is A_1 at time t_1 and A_2 at time t_2 ($t_2 > t_1$). Its mean life is T.

(A) $A_1t_1 = A_2t_2$ (B) $\frac{A_1 - A_2}{t_2 - t_1} = \text{constant}$

(C) $A_2 = A_1 e^{(t_1 - t_2)/T}$ (D) $A_2 = A_1 e^{(t_1/Tt_2)}$

MP0234

36. The decay constant of the end product of a radioactive series is

(A) zero (B) infinite

(C) finite (non zero) (D) depends on the end product.

MP0235

37. A radioactive substance is dissolved in a liquid and the solution is heated. The activity of the solution

(A) is smaller than that of element

(B) is greater than that of element

(C) is equal to that of element

(D) will be smaller or greater depending upon whether the solution is weak or concentrated.

MP0236

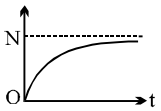
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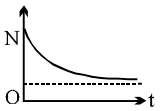
38. In a certain nuclear reactor, a radioactive nucleus is being produced at a constant rate = 1000 /s. The mean life of the radionuclide is 40 minutes. At steady state, the number of radionuclide will be

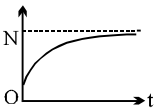
(A) 4×10^4 (B) 24×10^4 (C) 24×10^5 (D) 24×10^6

MP0237

39. In the above question, if there were 20×10^5 radionuclide at t = 0, then the graph of N v/s t is

(A) 

(B) 

(C) 

(D) 

MP0238

40. Which of the following cannot be emitted by radioactive substances during their decay ? **[AIEEE - 2003]**

(A) Protons (B) Neutrinos

(C) Helium nuclei (D) Electrons

MP0239

41. A 280 days old radioactive substance shows an activity of 6000 dps, 140 days later it's activity becomes 3000dps. What was its initial activity. **[JEE 2004 (Scr)]**

(A) 20000 dps (B) 24000 dps

(C) 12000 dps (D) 6000 dps

MP0240

42. The half-life period of a radioactive element X is same as the mean life time of another radioactive element Y. Initially they have the same number of atoms. Then- **[AIEEE - 2007]**

(A) X will decay faster than Y

(B) Y will decay faster than X

(C) Y and X have same decay rate initially

(D) X and Y decay at same rate always

MP0244

43. Given a sample of Radium-226 having half-life of 4 days. Find the probability, a nucleus disintegrates within 2 half lives. **[JEE 2006]**

(A) 1 (B) 1/2 (C) 3/4 (D) 1/4

MP0245

44. The half life of a radioactive substance is 20 minutes. The approximate time interval $(t_2 - t_1)$ between the time t_2 when $\frac{2}{3}$ of it has decayed and time t_1 when $\frac{1}{3}$ of it had decayed is :- **[AIEEE - 2011]**

(A) 20 min (B) 28 min (C) 7 min (D) 14 min

MP0281

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MATRIX MATCH TYPE QUESTION

B. α, β, γ, decay

45. Match the following Columns :- **[JEE 2006]**

	Column-I	**Column-II**
(A)	Nuclear fusion	(P) Converts some matter into energy
(B)	Nuclear fission	(Q) Generally occurs for nuclei with low atomic number
(C)	β–decay	(R) Generally occurs for nuclei with higher atomic number
(D)	Exothermic nuclear reaction	(S) Essentially proceeds by weak nuclear forces

MP0246

46. Some laws/processes are given in **Column I**. Match these with the physical phenomena given in **Column II** and indicate your answer by darkening appropriate bubbles in the 4 × 4 matrix given in the ORS. **[JEE 2007]**

Column-I	**Column-II**
(A) Transition between two atomic energy levels	(P) Characteristic X-rays
(B) Electron emission from a material	(Q) Photoelectric effect
(C) Mosley's law	(R) Hydrogen spectrum
(D) Change of photon energy into kinetic energy of electrons	(S) β-decay

MP0247

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EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

A. Binding energy of nucleus, nuclear fission & fusion

1. The binding energies of nuclei X and Y are E_1 and E_2 respectively. Two atoms of X fuse to give one atom of Y and an energy Q is released. Then:

(A) $Q = 2E_1 - E_2$ (B) $Q = E_2 - 2E_1$ (C) $Q = 2E_1 + E_2$ (D) $Q = 2E_2 + E_1$

MP0248

2. If each fission in a U^{235} nucleus releases 200 MeV, how many fissions must occurs per second to produce a power of 1 KW

(A) 1.325×10^{13} (B) 3.125×10^{13} (C) 1.235×10^{13} (D) 2.135×10^{13}

MP0249

3. The binding energies of the atom of elements A & B are E_a & E_b respectively. Three atoms of the element B fuse to give one atom of element A. This fusion process is accompanied by release of energy e. Then E_a, E_b are related to each other as

(A) $E_a + e = 3E_b$ (B) $E_a = 3E_b$ (C) $E_a - e = 3E_b$ (D) $E_a + 3E_b + e = 0$

MP0250

4. The rest mass of the deuteron, ${}_1^2H$, is equivalent to an energy of 1876 MeV, the rest mass of a proton is equivalent to 939 MeV and that of a neutron to 940 MeV. A deuteron may disintegrate to a proton and a neutron if it :

(A) emits a γ - ray photon of energy 2 MeV
(B) captures a γ - ray photon of energy 2 MeV
(C) emits a γ - ray photon of energy 3 MeV
(D) captures a γ - ray photon of energy 3 MeV

MP0251

5. In the nuclear fusion reaction,

$$ {}_1^2H + {}_1^3H \rightarrow {}_2^4He + n $$

given that the repulsive potential energy between the two nuclei is 7.7×10^{-14} J, the temperature at which the gases must be heated to initiate the reaction is nearly [Boltzmann's constant $k = 1.38 \times 10^{-23}$ J/K]- **[AIEEE - 2003]**

(A) 10^7 K (B) 10^5 K (C) 10^3 K (D) 10^9 K

MP0263

6. The binding energy per nucleon of deuteron $({}_1^2H)$ and helium nucleus $({}_2^4He)$ is 1.1 MeV and 7 MeV respectively. If two deuteron nuclei react to form a single helium nucleus, then the energy released is- **[AIEEE - 2004]**

(A) 13.9 MeV (B) 26.9 MeV (C) 23.6 MeV (D) 19.2 MeV

MP0264

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C. Radioactivity

7. In a radioactive element the fraction of initial amount remaining after its mean life time is

(A) $1-\frac{1}{e}$ (B) $\frac{1}{e^2}$ (C) $\frac{1}{e}$ (D) $1-\frac{1}{e^2}$

MP0253

8. Two radioactive material A_1 and A_2 have decay constants of $10\lambda_0$ and λ_0. If initially they have same number of nuclei, the ratio of number of their undecayed nuclei will be (1/e) after a time

(A) $\frac{1}{\lambda_0}$ (B) $\frac{1}{9\lambda_0}$ (C) $\frac{1}{10\lambda_0}$ (D) 1

MP0254

9. 90% of a radioactive sample is left undecayed after time t has elapsed. What percentage of the initial sample will decay in a total time 2t:

(A) 20% (B) 19% (C) 40% (D) 38%

MP0255

10. A radioactive material of half-life T was produced in a nuclear reactor at different instants, the quantity produced second time was twice of that produced first time. If now their present activities are A_1 and A_2 respectively then their age difference equals:

(A) $\frac{T}{\ln 2}\left|\ln\frac{A_1}{A_2}\right|$ (B) $T\left|\ln\frac{A_1}{A_2}\right|$ (C) $\frac{T}{\ln 2}\left|\ln\frac{A_2}{2A_1}\right|$ (D) $T\left|\ln\frac{A_2}{2A_1}\right|$

MP0256

11. The half-life of substance X is 45 years, and it decomposes to substance Y. A sample from a meteorite was taken which contained 2% of X and 14% of Y by quantity of substance. If substance Y is not normally found on a meteorite, what is the approximate age of the meteorite?

(A) 270 years (B) 135 years (C) 90 years (D) 45 years

MP0257

12. At time t = 0, N_1 nuclei of decay constant λ_1 & N_2 nuclei of decay constant λ_2 are mixed. The decay rate of the mixture is :

(A) $N_1N_2e^{-(\lambda_1+\lambda_2)t}$ (B) $+\left(\frac{N_1}{N_2}\right)e^{-(\lambda_1-\lambda_2)t}$

(C) $+\left(N_1\lambda_1e^{-\lambda_1t}+N_2\lambda_2e^{-\lambda_2t}\right)$ (D) $+N_1\lambda_1\,N_2\lambda_2\,e^{-(\lambda_1+\lambda_2)t}$

MP0258

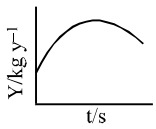
13. A radioactive nuclide can decay simultaneously by two different processes which have decay constants λ_1 and λ_2. The effective decay constant of the nuclide is λ, then :

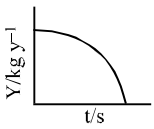
(A) $\lambda=\lambda_1+\lambda_2$ (B) $\lambda=1/2(\lambda_1+\lambda_1)$ (C) $\frac{1}{\lambda}=\frac{1}{\lambda_1}+\frac{1}{\lambda_2}$ (D) $\lambda=\sqrt{\lambda_1\lambda_2}$

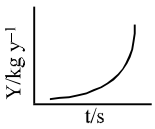
MP0259

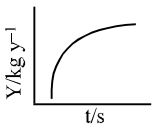
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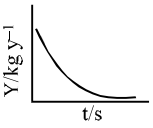
14. The radioactive nucleus of an element X decays to a stable nucleus of element Y. A graph of the rate of formation of Y against time would look like

(A) 

(B) 

(C) 

(D) 

(E) 

MP0260

15. The half life of a neutron is 800 sec. 10^8 neutrons at a certain instant are projected from one space station towards another space station, situated 3200 km away, with a velocity 2000 m/s. Their velocity remains constant during the journey. How many neutrons reach the other station?

(A) 50×10^6 (B) 25×10^6 (C) 80×10^5 (D) 25×10^5

MP0261

16. A radioactive source in the form of a metal sphere of diameter 3.2×10^{-3} m emits β-particle at a constant rate of 6.25×10^{10} particle/sec. The source is electrically insulated and all the β-particle are emitted from the surface. The potential of the sphere will rise to 1 V in time

(A) 180 μ sec (B) 90 μ sec (C) 18 μ sec (D) 9 μ sec

MP0262

MULTIPLE CORRECT TYPE QUESTIONS

A. Binding energy of nucleus, nuclear fission & fusion

17. When a nucleus with atomic number Z and mass number A undergoes a radioactive decay process:

(A) both Z and A will decrease, if the process is α decay

(B) Z will decrease but A will not change, if the process is β^+ decay

(C) Z will decrease but A will not change, if the process is β^- decay

(D) Z and A will remain unchanged, if the process is γ decay.

MP0265

18. When the atomic number A of the nucleus increases

(A) initially the neutron–proton ratio is constant = 1

(B) initially neutron–proton ratio increases and later decreases

(C) initially binding energy per nucleon increases and later decreases

(D) the binding energy per nucleon increases when the neutron–proton ratio increases.

MP0266

19. Let m_p be the mass of a proton, m_n the mass of a neutron, M_1 the mass of a $^{20}_{10}Ne$ nucleus and M_2 the mass of a $^{40}_{20}Ca$ nucleus. Then

(A) $M_2 = 2M_1$ (B) $M_2 > 2M_1$

(C) $M_2 < 2M_1$ (D) $M_1 < 10(m_n + m_p)$

MP0267

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20. Which of the following statement(s) is/are correct?

(A) The rest mass of a stable nucleus is less than the sum of the rest masses of its separated nucleons.

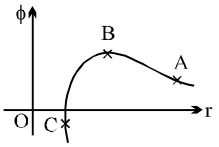
(B) The rest mass of a stable nucleus is greater than the sum of the rest masses of its separated nucleons.

(C) In nuclear fusion, energy is released by fusion of two nuclei of medium mass (approx. 100 amu)

(D) In nuclear fission, energy is released by fragmentation of a very heavy nucleus.

MP0269

21. The graph shown by the side shows the variation of potential energy ϕ of a proton with its distance 'r' from a fixed sodium nucleus, as it approaches the nucleus, placed at origin O. Then the portion.



(A) AB indicates nuclear repulsion

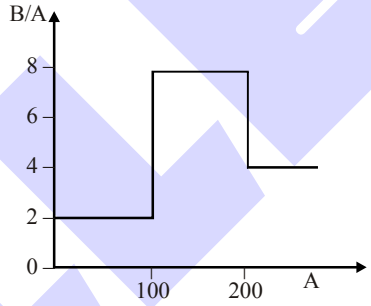
(B) AB indicates electrostatic repulsion

(C) BC indicates nuclear attraction

(D) BC represents electrostatic interaction

MP0270

22. Assume that the nuclear binding energy per nucleon (B/A) versus mass number (A) is as shown in the figure. Use this plot to choose the correct choice(s) given below : **[JEE 2008]**



(A) Fusion of two nuclei with mass numbers lying in the range of $1 < A < 50$ will release energy

(B) Fusion of two nuclei with mass numbers lying in the range of $51 < A < 100$ will release energy

(C) Fission of a nucleus lying in the mass range of $100 < A < 200$ will release energy when broken into two equal fragments

(D) Fission of a nucleus lying in the mass range of $200 < A < 260$ will release energy when broken into two equal fragments

MP0290

B. α, β, γ, decay

23. The instability of the nucleus can be due to various causes. An unstable nucleus emits radiations if possible to transform into less unstable state. Then the cause and the result can be

(A) a nucleus of excess nucleons is α active

(B) an excited nucleus of excess protons is β^- active

(C) an excited nucleus of excess protons is β^+ active

(D) an nucleus of excess neutrons is β^- active

MP0271

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24. In β-decay, the Q-value of the process is E. Then
(A) K.E. of a β-particle cannot exceed E.
(B) K.E. of anti neutrino emitted lies between Zero and E.
(C) N/Z ratio of the nucleus is altered.
(D) Mass number (A) of the nucleus is altered.

MP0272

25. Consider the following nuclear reactions and select the correct statements from the options that follow.
Reaction I : $n \rightarrow p + e^- + \bar{\nu}$ **Reaction II :** $p \rightarrow n + e^+ + \nu$
(A) Free neutron is unstable, therefore reaction I is possible
(B) Free proton is stable, therefore reaction II is not possible
(C) Inside a nucleus, both decays (reaction I and II) are possible
(D) Inside a nucleus, reaction I is not possible but reaction II is possible.

MP0273

26. When the nucleus of an electrically neutral atom undergoes a radioactive decay process, it will remain neutral after the decay if the process is :
(A) α decay (B) β^- decay (C) γ decay (D) K-capture

MP0274

C. Radioactivity

27. The decay constant of a radio active substance is 0.173 $(\text{years})^{-1}$. Therefore :
(A) Nearly 63% of the radioactive substance will decay in (1/0.173) year.
(B) half life of the radio active substance is (1/0.173) year.
(C) one -forth of the radioactive substance will be left after nearly 8 years.
(D) all the above statements are true.

MP0268

COMPREHENSION TYPE QUESTIONS

B. α, β, γ, decay

Paragraph for Questions 28 and 29

The β-decay process, discovered around 1900, is basically the decay of a neutron (n). In the laboratory, a proton (p) and an electron (e^-) are observed as the decay products of the neutron. Therefore, considering the decay of a neutron as a two-body decay process, it was predicted theoretically that the kinetic energy of the electron should be a constant. But experimentally, it was observed that the electron kinetic energy has a continuous spectrum. Considering a three-body decay process, i.e. $n \rightarrow p + e^- + \vec{\nu}_e$, around 1930, Pauli explained the observed electron energy spectrum. Assuming the anti-neutrino $(\vec{\nu}_e)$ to be massless and possessing negligible energy, and the neutron to be at rest, momentum and energy conservation principles are applied. From this calculation, the maximum kinetic energy of the electron is 0.8×10^6 eV. The kinetic energy carried by the proton is only the recoil energy.

28. If the anti-neutrino had a mass of 3 eV/c^2 (where c is the speed of light) instead of zero mass, what should be the range of the kinetic energy, K, of the electron? **[JEE 2012]**

(A) $0 \le K \le 0.8 \times 10^6 eV$ (B) $3.0eV \le K \le 0.8 \times 10^6 eV$
(C) $3.0eV \le K < 0.8 \times 10^6 eV$ (D) $0 \le K < 0.8 \times 10^6 eV$

MP0295

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29. What is the maximum energy of the anti-neutrino? **[JEE 2012]**

(A) zero

(B) much less than 0.8×10^6 eV

(C) Nearly 0.8×10^6 eV

(D) Much larger than 0.8×10^6 eV

MP0295

MATRIX MATCH TYPE QUESTION

A. Binding energy of nucleus, nuclear fission & fusion

30. In the following, column I lists some physical quantities & the column II gives approx. energy values associated with some of them. Choose the appropriate value of energy from column II for each of the physical quantities in column I and write the corresponding letter A, B, C etc. against the number (i), (ii), (iii), etc. of the physical quantity in the answer book. In your answer, the sequence of column I should be maintained.

	Column-I		**Column-II**
(i)	Energy of thermal neutrons	(A)	0.025 eV
(ii)	Energy of X-rays	(B)	0.5 eV
(iii)	Binding energy per nucleon	(C)	3 eV
(iv)	Photoelectric threshold of metal	(D)	20 eV
		(E)	10 keV
		(F)	8 MeV

MP0275

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EXERCISE (S)

A. Binding energy of nucleus, nuclear fission & fusion

1. Suppose that the Sun consists entirely of hydrogen atom and releases the energy by the nuclear reaction, $4\,{}_1^1H \longrightarrow {}_2^4He$ with 26 MeV of energy released. If the total output power of the Sun is assumed to remain constant at 3.9×10^{26} W, find the time it will take to burn all the hydrogen. Take the mass of the Sun as 1.7×10^{30} kg.

MP0190

2. When two deuterons ($_1H^2$) fuse to from a helium nucleus $_2He^4$, 23.6 MeV energy is released. Find the binding energy of helium if it is 1.1 MeV for each nucleon of deuterium.

MP0191

3. Highly energetic electrons are bombarded on a target of an element containing 30 neutrons. The ratio of radii of nucleus to that of helium nucleus is $(14)^{1/3}$. Find **[JEE 2005]**

(a) atomic number of the nucleus

(b) the frequency of K_α line of the X-ray produced. ($R = 1.1 \times 10^7\ m^{-1}$ and $c = 3 \times 10^8$ m/s)

MP0192

B. α, β, γ, decay

4. The kinetic energy of an $\alpha-$particle which flies out of the nucleus of a Ra^{226} atom in radioactive disintegration is 4.78 MeV. Find the total energy evolved during the escape of the $\alpha-$particle.

MP0193

C. Radioactivity

5. U^{238} and U^{235} occur in nature in an atomic ratio 140 : 1. Assuming that at the time of earth's formation the two isotopes were present in equal amounts. Calculate the age of the earth.

(Half life of $u^{238} = 4.5 \times 10^9$ yrs & that of $U^{235} = 7.13 \times 10^8$ yrs)

MP0194

6. At t = 0, a sample is placed in a reactor. An unstable nuclide is produced at a constant rate R in the sample by neutron absorption. This nuclide β^- decays with half life τ. Find the time required to produce 80% of the equilibrium quantity of this unstable nuclide.

MP0195

7. The activity of a freshly prepared radioactive sample is 10^{10} disintegrations per second, whose mean life is 10^9 s. The mass of an atom of this radioisotope is 10^{-25} kg. The mass (in mg) of the radioactive sample is **[JEE-2011]**

MP0294

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EXERCISE - JM

1. Half-lives of two radioactive elements A and B are 20 minutes and 40 minutes, respectively. Initially, the samples have equal number of nuclei. After 80 minutes, the ratio of decayed numbers of A and B nuclei will be :- **[JEE-Main-2016]**

(1) 5 : 4 (2) 1 : 16 (3) 4 : 1 (4) 1 : 4

MP0287

2. A radioactive nucleus A with a half life T, decays into a nucleus B. At t = 0, there is no nucleus B. At sometime t, the ratio of the number of B to that of A is 0.3. Then, t is given by : **[JEE-Main-2017]**

(1) $t = T \log (1.3)$ (2) $t = \frac{T}{\log(1.3)}$ (3) $t = \frac{T}{2}\frac{\log 2}{\log 1.3}$ (4) $t = T\frac{\log 1.3}{\log 2}$

MP0288

SELECTED PROBLEMS FROM JEE-MAINS ONLINE PAPERS

3. Two radioactive substances A and B have decay constants 5λ and λ respectively. At t = 0, a sample has the same number of the two nuclei. The time taken for the ratio of the number of nuclei to become $\left(\frac{1}{e}\right)^2$ will be : **[JEE Main-2019_April]**

(1) $1 / 4\lambda$ (2) $1 / \lambda$ (3) $1 / 2\lambda$ (4) $2 / \lambda$

MP0379

4. Half lives of two radioactive nuclei A and B are 10 minutes and 20 minutes, respectively. If, initially a sample has equal number of nuclei, then after 60 minutes, the ratio of decayed numbers of nuclei A and B will be : **[JEE Main-2019_April]**

(1) 9 : 8 (2) 1 : 8 (3) 8 : 1 (4) 3 : 8

MP0380

5. In a reactor, 2 kg of ${}_{92}U^{235}$ fuel is fully used up in 30 days. The energy released per fission is 200 MeV. Given that the Avogadro number, $N = 6.023 \times 10^{26}$ per kilo mole and 1 eV = 1.6×10^{-19} J. The power output of the reactor is close to : **[JEE Main-2020_Sep]**

(1) 125 MW (2) 60 MW (3) 35 MW (4) 54 MW

MP0381

6. You are given that Mass of ${}^{7}_{3}Li$ = 7.0160 u,

Mass of ${}^{4}_{2}He$ = 4.0026 u

and Mass of ${}^{1}_{1}H$ = 1.0079 u.

When 20 g of ${}^{7}_{3}Li$ is converted into ${}^{4}_{2}He$ by proton capture, the energy liberated, (in kWh), is: [Mass of nudeon = 1 GeV/c^2] **[JEE Main-2020_Sep]**

(1) 8×10^6 (2) 1.33×10^6 (3) 6.82×10^5 (4) 4.5×10^5

MP0383

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7. Given the masses of various atomic particles $m_p = 1.0072u$, $m_n = 1.0087u$, $m_e = 0.000548u$, $m_{\bar{\nu}} = 0$, $m_d = 2.0141u$, where p ≡ proton, n ≡ neutron, e ≡ electron, $\bar{\nu}$ ≡ antineutrino and d ≡ deuteron. Which of the following process is allowed by momentum and energy conservation ? **[JEE Main-2020_Sep]**

(1) $n + p \rightarrow d + \gamma$

(2) $e^+ + e^- \rightarrow \gamma$

(3) n + n → deuterium atom (electron bound to the nucleus)

(4) $p \rightarrow n + e^+ + \bar{\nu}$

MP0384

8. Find the bindng energy per nucleon for $^{120}_{50}Sn$. Mass of proton m_p = 1.00783 U, mass of neutron m_n = 1.00867 U and mass of tin nucleus m_{Sn} = 119.902199 U. (take 1U = 931 MeV)

[JEE Main-2020_Sep]

(1) 8.5 MeV (2) 7.5 MeV (3) 8.0 MeV (4) 9.0 MeV

MP0385

9. The half-life of ^{198}Au is 3 days. If atomic weight of ^{198}Au is 198 g/mol then the activity of 2 mg of ^{198}Au is [in disintegration/second] : **[JEE Main-2021_July]**

(1) 2.67×10^{12} (2) 6.06×10^{18} (3) 32.36×10^{12} (4) 16.18×10^{12}

MP0398

10. A radioactive sample has an average life of 30 ms and is decaying. A capacitor of capacitance 200 F is first charged and later connected with resistor 'R'. If the ratio of charge on capacitor to the activity of radioactive sample is fixed with respect to time then the value of 'R' should be _____ Ω.

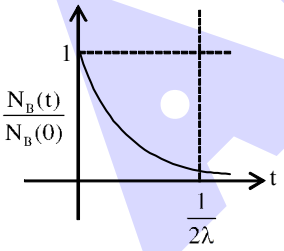
[JEE Main-2021_July]

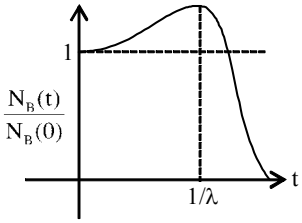
MP0399

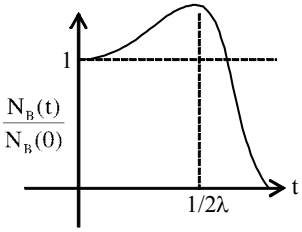
11. At time t = 0, a material is composed of two radioactive atoms A and B, where $N_A(0) = 2N_B(0)$. The decay constant of both kind of radioactive atoms is λ. However, A disintegrates to B and B disintegrates to C. Which of the following figures represents the evolution of $N_B(t) / N_B(0)$ with respect to time t ?

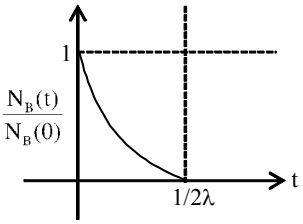
$$\begin{bmatrix} N_A(0) = \text{No. of A atoms at } t = 0 \\ N_B(0) = \text{No. of B atoms at } t = 0 \end{bmatrix}$$

[JEE Main-2021_July]

(1) 

(2) 

(3) 

(4) 

MP0400

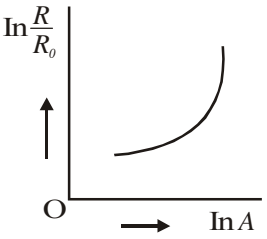
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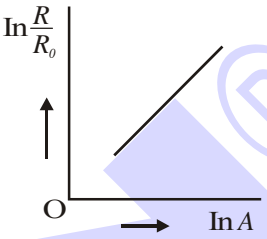
12. Nucleus A is having mass number 220 and its binding energy per nucleon is 5.6 MeV. It splits in two fragments 'B' and 'C' of mass numbers 105 and 115. The binding energy of nucleons in 'B' and 'C' is 6.4 MeV per nucleon. The energy Q released per fission will be : **[JEE Main-2022_June]**

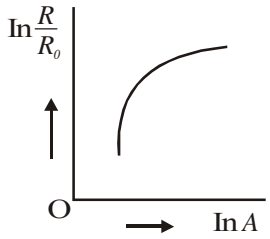
(A) 0.8 MeV (B) 275 MeV (C) 220 MeV (D) 176 MeV

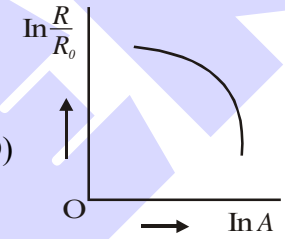
MP0421

13. Which of the following figure represents the variation of $\ln\left(\frac{R}{R_0}\right)$ with $\ln A$(If R = radius of a nucleus and A = its mass number) **[JEE Main-2022_June]**

(A) 

(B) 

(C) 

(D) 

MP0422

14. Following statements related to radioactivity are given below: **[JEE Main-2022_June]**

(A) Radioactivity is a random and spontaneous process and is dependent on physical and chemical conditions.

(B) The number of un-decayed nuclei in the radioactive sample decays exponentially with time.

(C) Slope of the graph of loge(no. of undecayed nuclei) Vs. time represents the reciprocal of mean life time (τ).

(D) Product of decay constant (λ) and half-life time ($T_{1/2}$) is not constant.

Choose the most appropriate answer from the options given below:

(A)(A) and (B) only (B) (B) and (D) only

(C) (B) and (C) only (D) (C) and (D) only **MP0423**

15. The Q-value of a nuclear reaction and kinetic energy of the projectile particle, K_p are related as : **[JEE Main-2022_June]**

(A) $Q = K_p$ (B) $(K_p + Q) < O$ (C) $Q < K_p$ (D) $(K_p + Q) > 0$

MP0424

16. The activity of a radioactive material is 2.56×10^{-3} Ci. If the half life of the material is 5 days, after how many days the activity will become 2×10^{-5} Ci? **[JEE Main-2022_June]**

(A) 30 days (B) 35 days (C) 40 days (D) 25 days

MP0425

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17. The disintegration rate of a certain radioactive sample at any instant is 4250 disintegrations per minute. 10 minutes later, the rate becomes 2250 disintegrations per minute. The approximate decay constant is : (Take $\log_{10} 1.88 = 0.274$) **[JEE Main-2022_July]**

(A) 0.02 min^{-1} (B) 2.7 min^{-1} (C) 0.063 min^{-1} (D) 6.3 min^{-1}

MP0426

18. Two lighter nuclei combine to form a comparatively heavier nucleus by the relation given below:

$${}_1^2X + {}_1^2X = {}_2^4Y$$

The binding energies per nucleon ${}_1^2X$ and ${}_2^4Y$ are 1.1 MeV and 7.6 MeV respectively. The energy released in this process is_____ . MeV. **[JEE Main-2022_July]**

MP0427

19. What is the half-life period of a radioactive material if its activity drops to $1/16^{th}$ of its initial value of 30 years ? **[JEE Main-2022_July]**

(A) 9.5 years (B) 8.5 years (C) 7.5 years (D) 10.5 years

MP0428

20. The activity of a radioactive material is 6.4×10^{-4} curie. Its half life is 5 days. The activity will become 5×10^{-6} curie after : **[JEE Main-2022_July]**

(A) 7 days (B) 15 days (C) 25 days (D) 35 days

MP0429

21. A freshly prepared radioactive source of half life 2 hours 30 minutes emits radiation which is 64 times the permissible safe level. The minimum time, after which it would be possible to work safely with source, will be _____ hours. **[JEE Main-2022_July]**

MP0430

22. A radioactive sample decays $\frac{7}{4}$ times its original quantity in 15 minutes. The half-life of the sample is

[JEE Main-2022_July]

(A) 5 min (B) 7.5 min (C) 15 min (D) 30 min

MP0431

23. Read the following statements:

(A) Volume of the nucleus is directly proportional to the mass number.

(B) Volume of the nucleus is independent of mass number.

(C) Density of the nucleus is directly proportional to the mass number.

(D) Density of the nucleus is directly proportional to the cube root of the mass number.

(E) Density of the nucleus is independent of the mass number.

Choose the correct option from the following options. **[JEE Main-2022_July]**

(A) (A) and (D) only. (B) (A) and (E) only.

(C) (B) and (E) only. (D) (A) and (C) only

MP0432

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EXERCISE - JA

1. A freshly prepared sample of a radioisotope of half-life 1386 s has activity 10^3 distintegrations per second. Given that $\ell n\ 2 = 0.693$, the fraction of the initial number of nuclei (expressed in nearest integer percentage) that will decay in the first 80 s after preparation of the sample is.

[JEE Advance-2013]

MP0296

Paragraph for Questions 2 and 3

The mass of a nucleus ${}^{A}_{Z}X$ is less than the sum of the masses of (A – Z) number of neutrons and Z number of protons in the nucleus. The energy equivalent to the corresponding mass difference is known as the binding energy of the nucleus. A heavy nucleus of mass M can break into two light nuclei of masses m_1 and m_2 only if $(m_1 + m_2) < M$. Also two light nuclei of masses m_3 and m_4 can undergo complete fusion and form a heavy nucleus of mass M' only if $(m_3 + m_4) > M'$. The masses of some neutral atoms are given in the table below :- **[JEE Advance-2013]**

${}^{1}_{1}H$	1.007825 u	${}^{2}_{1}H$	2.014102 u	${}^{3}_{1}H$	3.016050 u	${}^{4}_{2}He$	4.002603 u
${}^{6}_{3}Li$	6.015123 u	${}^{7}_{3}Li$	7.016004 u	${}^{70}_{30}Zn$	69.925325 u	${}^{82}_{34}Se$	81.916709 u
${}^{152}_{64}Gd$	151.919803 u	${}^{206}_{82}Pb$	205.974455 u	${}^{209}_{83}Bi$	208.980388 u	${}^{210}_{84}Po$	209.982876 u

$(1u = 932\ MeV/c^2)$

2. The kinetic energy (in keV) of the alpha particle, when the nucleus ${}^{210}_{84}Po$ at rest undergoes alpha decay, is :-

(A) 5319 (B) 5422

(C) 5707 (D) 5818

MP0297

3. The correct statement is :-

(A) The nucleus ${}^{6}_{3}Li$ can emit an alpha particle

(B) The nucleus ${}^{210}_{84}Po$ can emit a proton

(C) Deuteron and alpha particle can undergo complete fusion

(D) The nuclei ${}^{70}_{30}Zn$ and ${}^{82}_{34}Se$ can undergo complete fusion

MP0297

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4. Match List I of the nuclear processes with List II containing parent nucleus and one of the end products of each process and then select the correct answer using the codes given below the lists:

[JEE Advance-2013]

	List-I		**List-II**
P.	Alpha decay	1.	$^{15}_{8}O \rightarrow ^{15}_{7}N + ...$
Q.	β^+ decay	2.	$^{238}_{92}U \rightarrow ^{234}_{90}Th + ...$
R.	Fission	3.	$^{185}_{83}Bi \rightarrow ^{184}_{82}Pb + ...$
S.	Proton emission	4.	$^{239}_{94}Pu \rightarrow ^{140}_{57}La + ...$

Codes :

	P	Q	R	S
(A)	4	2	1	3
(B)	1	3	2	4
(C)	2	1	4	3
(D)	4	3	2	1

MP0298

5. A nuclear power plant supplying electrical power to a village uses a radioactive material of half life T years as the fuel. The amount of fuel at the beginning is such that the total power requirement of the village is 12.5% of the electrical power available from the plant at that time. If the plant is able to meet the total power needs of the village for a maximum period of nT years, then the value of n is.

[JEE Advance-2015]

MP0301

6. Match the nuclear processes given in column-I with the appropriate option(s) in column-II.

[JEE Advance-2015]

	Column-I		**Column-II**
(A)	Nuclear fusion	(P)	Absorption of thermal neutrons by $^{235}_{92}U$
(B)	Fission in a nuclear reactor	(Q)	$^{60}_{27}Co$ nucleus
(C)	β-decay	(R)	Energy production in stars via hydrogen conversion to helium
(D)	γ-ray emission	(S)	Heavy water
		(T)	Neutrino emission

MP0302

7. For a radioactive material, its activity A and rate of change of its activity R are defined as $A = -\frac{dN}{dt}$ and $R = -\frac{dA}{dt}$, where N(t) is the number of nuclei at time t. Two radioactive sources P (mean life τ) and Q (mean life 2τ) have the same activity at t = 0. Their rates of change of activities at t = 2τ are R_P and R_Q, respectively. If $\frac{R_P}{R_Q} = \frac{n}{e}$, then the value of n is : **[JEE Advance-2015]**

MP0303

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8. A fission reaction is given by $^{236}_{92}U \rightarrow ^{140}_{54}Xe + ^{94}_{38}Sr + x + y$, where x and y are two particles. Considering $^{236}_{92}U$ to be at rest, the kinetic energies of the products are denoted by $K_{Xe}, K_{sr}, K_x(2MeV)$ and $K_y(2MeV)$, respectively. Let the binding energies per nucleon of $^{236}_{92}U$, $^{140}_{54}Xe$ and $^{94}_{38}Sr$ be 7.5 MeV, 8.5 MeV and 8.5 MeV, respectively. Considering different conservation laws, the correct options(s) is (are) :- **[JEE Advance-2015]**

(A) x = n, y = n, K_{sr} = 129 MeV, K_{xe} = 86 MeV

(B) x = p, y = e^-, K_{Sr} = 129 MeV, K_{xe} = 86 MeV

(C) x = p, y = n, K_{Sr} = 129 MeV, K_{xe} = 86 MeV

(D) x = n, y = n, K_{Sr} = 86 MeV, K_{xe} = 129 MeV

MP0304

9. The isotope $^{12}_{5}B$ having a mass 12.014 u undergoes β-decay to $^{12}_{6}C$. $^{12}_{6}C$ has an excited state of the nucleus ($^{12}_{6}C^*$) at 4.041 MeV above its ground state. If $^{12}_{5}B$ decays to $^{12}_{6}C^*$, the maximum kinetic energy of the β-particle in units of MeV is (1u = 931.5 MeV/c^2, where c is the speed of light in vacuum). **[JEE Advance-2016]**

MP0305

10. The electrostatic energy of Z protons uniformly distributed throughout a spherical nucleus of radius R is given by $E = \frac{3}{5}\frac{Z(Z-1)e^2}{4\pi\varepsilon_0 R}$ **[JEE Advance-2016]**

The measured masses of the neutron, $^{1}_{1}H$, $^{15}_{7}N$ and $^{15}_{8}O$ are 1.008665 u, 1.007825 u, 15.000109 u and 15.003065 u respectively. Given that the radii of both the $^{15}_{7}N$ and $^{15}_{8}O$ nuclei are same, 1u = 931.5 MeV/c^2 (c is the speed of light) and $\frac{e^2}{(4\pi\varepsilon_0)}$ = 1.44 MeV fm. Assuming that the difference between the binding energies of $^{15}_{7}N$ and $^{15}_{8}O$ is purely due to the electrostatic energy, the radius of either of the nuclei is (1fm = 10^{-15}m)

(A) 2.85 fm (B) 3.03 fm (C) 3.42 fm (D) 3.80 fm

MP0306

11. An accident in a nuclear laboratory resulted in deposition of a certain amount of radioactive material of half-life 18 days inside the laboratory. Tests revealed that the radiation was 64 times more than the permissible level required for safe operation of the laboratory. What is the minimum number of days after which the laboratory can be considered safe for use? **[JEE Advance-2016]**

(A) 64 (B) 90 (C) 108 (D) 120

MP0307

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12. ^{131}I is an isotope of Iodine that β decays to an isotope of Xenon with a half-life of 8 days. A small amount of a serum labelled with ^{131}I is injected into the blood of a person. The activity of the amount of ^{131}I injected was 2.4×10^5 Becquerel (Bq). It is known that the injected serum will get distributed uniformly in the blood stream in less than half an hour. After 11.5 hours, 2.5 ml of blood is drawn from the person's body, and gives an activity of 115 Bq. The total volume of blood in the person's body, in liters is approximately (you may use $e^x \approx 1 + x$ for $|x| << 1$ and $\ln 2 \approx 0.7$).

[JEE Advance-2017]

MP0308

13. In a radioactive decay chain, $^{232}_{90}Th$ nucleus decays to $^{212}_{82}Pb$ nucleus. Let N_α and N_β be the number of α and β^- particles, respectively, emitted in this decay process. Which of the following statements is (are) true ? **[JEE Advance-2018]**

(A) $N_\alpha = 5$ (B) $N_\alpha = 6$ (C) $N_\beta = 2$ (D) $N_\beta = 4$

MP0309

14. Suppose a $^{226}_{88}Ra$ nucleus at rest and in ground state undergoes α-decay to a $^{222}_{86}Rn$ nucleus in its excited state. The kinetic energy of the emitted α particle is found to be 4.44 MeV. $^{222}_{86}Rn$ nucleus then goes to its ground state by γ-decay. The energy of the emitted γ-photon is _____ keV,

[Given: atomic mass of $^{226}_{88}Ra = 226.005u$, atomic mass of $^{222}_{86}Rn = 222.000u$, atomic mass of α particle = 4.000u, 1u = 931 MeV/c^2, c is speed of the light] **[JEE Advanced-2019]**

MP0310

15. A heavy nucleus Q of half-life 20 minutes undergoes alpha-decay with probability of 60% and beta-decay with probability of 40%. Initially, the number of Q nuclei is 1000. The number of alpha-decays of Q in the first one hour is **[JEE Advanced-2021]**

(A) 50 (B) 75 (C) 350 (D) 525

MP0401

16. A heavy nucleus N, at rest, undergoes fission $N \rightarrow P + Q$, where P and Q are two lighter nuclei. Let $\delta = M_N - M_P - M_Q$, where M_P, M_Q and M_N are the masses of P, Q and N, respectively. E_P and E_Q are the kinetic energies of P and Q, respectively. The speed of P and Q are v_P and v_Q, respectively. If c is the speed of light, which of the following statement(s) is(are) correct ? **[JEE Advanced-2021]**

(A) $E_P + E_Q = c^2\delta$

(B) $E_P = \left(\frac{M_P}{M_P + M_Q}\right) c^2\delta$

(C) $\frac{v_P}{v_Q} = \frac{M_Q}{M_P}$

(D) The magnitude of momentum for P as well as Q is $c\sqrt{2\mu\delta}$, where $\mu = \frac{M_P M_Q}{(M_P + M_Q)}$

MP0402

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17. The minimum kinetic energy needed by an alpha particle to cause the nuclear reaction ${}^{16}_{7}\mathrm{N} + {}^{4}_{2}\mathrm{He} \rightarrow {}^{1}_{1}\mathrm{H} + {}^{19}_{8}\mathrm{O}$ in a laboratory frame is n (in MeV). Assume that ${}^{16}_{7}\mathrm{N}$ is at rest in the laboratory frame. The masses of ${}^{16}_{7}\mathrm{N}$, ${}^{4}_{2}\mathrm{He}$, ${}^{1}_{1}\mathrm{H}$ and ${}^{19}_{8}\mathrm{O}$ can be taken to be 16.006 u, 4.003 u, 1.008 u and 19.003 u, respectively, where 1 u = 930 MeVc^{-2}. The value of n is_____.

[JEE Advanced-2022]

MP0433

18. The binding energy of nucleons in a nucleus can be affected by the pairwise Coulomb repulsion. Assume that all nucleons are uniformly distributed inside the nucleus. Let the binding energy of a proton be E_b^p and the binding energy of a neutron be E_b^n in the nucleus.

Which of the following statement(s) is(are) CORRECT? **[JEE Advanced-2022]**

(A) $E_b^p - E_b^n$ is proportional to Z(Z – 1) where Z is the atomic number of the nucleus.

(B) $E_b^p - E_b^n$ is proportional to $A^{-\frac{1}{3}}$ where A is the mass number of the nucleus.

(C) $E_b^p - E_b^n$ is positive.

(D) E_b^p increases if the nucleus undergoes a beta decay emitting a positron.

MP0434

19. In a radioactive decay chain reaction, ${}^{230}_{90}\mathrm{Th}$ nucleus decays into ${}^{214}_{84}\mathrm{Po}$ nucleus. The ratio of the number of α to number of β^- particles emitted in this process is _____. **[JEE Advanced-2022]**

MP0435

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ANSWER KEY

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	C	A	A	D	A	B	A	B	D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	C	B	C	B	A	C	D	B	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	A	B	C	B	C	D	D	D	D
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	C	B	B	B	C	A	C	C	C	A
Que.	41	42	43	44	45					
Ans.	B	B	C	A	(A)-P,Q; (B)-P,R; (C)-S,P; (D)-P,Q,R					
Que.	46									
Ans.	(A)-R, P; (B)-Q, S; (C)-P; (D)-Q									

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	B	C	D	D	C	C	B	B	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	C	A	E	B	C	A,B,D	A,C	C,D	A,D
Que.	21	22	23	24	25	26	27	28	29	
Ans.	B,C	B,D	A,C,D	A,B,C	A,B,C	C,D	A,C	D	C	
Que.	30									
Ans.	(i)-A; (ii)-E; (iii)-F; (iv)-C									

EXERCISE # S

1. $8/3\times10^{18}$ sec **2.** 28 MeV **3.** v = 1.546 × 10^{18} Hz; Z = 26 **4.** 4.87 MeV

5. 6.04 × 10^{9} yrs **6.** $t=\left(\frac{\ln 5}{\ln 2}\right)\tau$ **7.** 1

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EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	4	3	1	2	2	1	1	4	150
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	D	B	C	D	B	C	26	C	D
Que.	21	22	23							
Ans.	15	A	B							

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6				
Ans.	4	A	C	C	3	(A)-R or R,T; (B)-P,S; (C)-Q,T; (D)-R				
Que.	7	8	9	10	11	12	13	14	15	16
Ans.	2	A	9	C	C	5	A,C	135.00	D	A,C,D
Que.	17	18	19							
Ans.	2.32 to 2.33	A,B,D	2							

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